Absence Is Not Omission: Availability, Vintage, and the Measurement of Selective Benchmark Disclosure

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Abstract

Providers choose which benchmark results a release document prints, while independent evaluators score the same models on many more. Dating the instruments seems to settle what could have been reported, since a benchmark published after a model shipped could not be. The comparison fails on its own terms: the date that decides availability also fixes where a model sits in each benchmark's comparison pool, so benchmarks that postdate a release already trail those that predate it before any label enters. The failure is visible on a leaderboard we did not build, where fixed models keep identical scores across fourteen HELM Lite releases while every published win rate moves, and a companion leaderboard with an absolute headline moves nothing. Hand-coding every locatable release document shows the consequence: the coded gap rejects zero at +11.4 percentile points, yet removing every label returns +14.1 on the same releases. A study built this way turns pool composition into apparent concealment. What the record does establish is composition, reported benchmarks standing 8.7 points above omitted eligible ones within release, while no possession-grounded test detects withholding and the change-based drop gap keeps its predicted sign inside the permutation null. Carrying a fixed suite across releases would make later silence informative; adding labels to a moving pool does not.

1 Introduction

For each frontier model release, providers disclose benchmark scores through a model card, while third parties execute standardised benchmarks for the same models [Liang et al., 2023, Epoch AI, 2026]. This paper asks whether benchmark availability can identify the older economic question of whether the provider's undisclosed scores are typically worse than their reported scores. Only when the signal is common knowledge does disclosure unravel [Grossman, 1981, Milgrom, 1981]. As Dye [1985] details, when receivers cannot distinguish between senders who have no information, and senders who have bad information, receivers treat senders as the same, encouraging withholding. This indistinguishability extends to any research that attempts to analyse what was withheld, since we rely on the endowment, the signal a sender had at time of writing, which we do not observe in the work below.

The presence of benchmark release dates gives a rare ability to bound the endowment exogenously: when benchmarks are never shared before publication, any benchmark released after the model release is necessarily not in the provider's endowment, regardless of whether providers ran it themselves. This lets us distinguish what was able to be disclosed, from what was unable to be disclosed.

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First, we provide a new longitudinal study of benchmark disclosure that uses the boundary to construct a panel of a fixed snapshot of an independent benchmarking hub containing 819 model-versions, which we aggregate over scaffold variants to a panel of 353 provider releases, 46 organisations, 74 benchmarks, and 3,082 model-benchmark scores. We provide a verified release date for each benchmark, creating a new panel of 2,355 eligible model-benchmark scores, and 476 ineligible model-benchmark scores.

Second, we show that while the ability to bound the endowment is a unique and powerful feature to understand withholding, constructing a relative ranking of performance is required to judge withholding. Our central analysis compares reported, versus omitted benchmarks that were able to be reported, and requires the relative ranking to be unbiased across the two sides of availability. Instead, there is a 12.23 percentile point difference on average in performance between benchmarks created before the model release, versus those that were created after the model release, with pre-release benchmarks outperforming post-release benchmarks in 78.1% of the 146 releases that have cells in both groups. None of this calculation uses whether the provider reported the benchmark, and the least biased estimate of the relative ranking remains at 9.14 points. This remains a structural problem, as the verification of the endowment uses the same signal, the release date, that biases our relative ranking [Heckman, 1979]. In economic terms, the presence of benchmark vintages is a free signal to verify the endowment, which Dye [1985] demonstrates cannot be known to receivers in general. However, the signal to verify the endowment is not excludable in the relative ranking, and therefore is self-contaminating. This raises a fundamental design challenge for any research that verifies the presence of information via vintage, but then prices evidence using a relative ranking, win rate, or percentile. Not only must reporting decisions be observable, but also the relative ranking must maintain a consistent reference group that is invariant to the endowment.

Third, we extend these concerns beyond our panel. In a sample of 14 published releases for HELM Lite, the pool expands (30 → 91 models); 24 models keep bit-for-bit identical scores; and all their published headline values change (five pairs reversing order). These changes do not show up in a parallel leaderboard maintained by the same group of authors, which presents an absolute headline.

Fourth, we code this history. We hand-code the 100 releases with a locatable release-time document, independently replicated on a fifth ($\kappa = 0.95$ on reported vs. not), yielding the estimator such a study would use. Omitted benchmarks are +11.4 points higher than those the provider could not have reported (excluding zero, but falling short of its matched counterpart by what composition implies: it cannot separate concealment from the availability gap). What survives is composition: reported benchmarks are 8.7 points higher than omitted eligible benchmarks within release ($p = 0.0001$), robust to benchmark fixed effects.

1.1 Related literature

The state of reporting in model documentation is imperfect, incomplete, and dependent on the sender, established [Staufer et al., 2025, Kuehnert et al., 2026], with two papers focusing on the reporting gaps themselves: Zhang et al. [2026] finds a benchmark family missing from all main result tables across four flagship releases, and Ghosh et al. [2026] defines the reporting standard that would make the omission observable. Both lack benchmark dates, so there's no way to distinguish a skipped benchmark from one the release predates. Bean et al. [2025] instead evaluates 445 benchmarks for construct validity, a different question about whether instruments measure what they're claimed to, while we take the instruments as given.

Evidence is stronger on leaderboards: Singh et al. [2025] characterises selection in real time (providers testing private variants and publishing the best), while Xu et al. [2025] shifted HELM's primary metric off a win rate that depends on the models compared. In both cases, the leaderboard changes for reasons other than the evidence, but neither isolates, and we do at the cell level, an omission from a result that could not have existed. What no paper here controls for is availability, which we introduce: absent a confirmed benchmark date, the distinction between "wasn't reported" and "did not exist" collapses.

The economics literature frames the problem and gives a cautionary note: silence unravels when endowments are commonly observed [Grossman, 1981, Milgrom, 1981] but persists if the recipient cannot distinguish an empty endowment from a concealed one [Dye, 1985, Jung and Kwon, 1988] or when revelation incurs a cost [Verrecchia, 1983], frictions our design does not separate. Empirically, the warning bears out: unravelling doesn't occur in practice [Dranove and Jin, 2010, Jin and Leslie, 2003, Hotz and Xiao, 2013, Bederson et al., 2018] and in the lab [Jin et al., 2021]. What all this work misses is the endowment directly, the thing our bound aims to recover. Meanwhile, concurrent theory takes a direct approach to strategic assessment and persuasion [Truong et al., 2026, Dai et al., 2026]; theory, not measurement.

2 Data and Setting

2.1 The disclosure problem

To begin, we assume release i occurs at time t_i and contains a single document. Benchmarks are indexed by $b \in \mathcal{B}$. They are released at time τ_b. At the time of the writing of release i the provider is endowed with results for a set of benchmarks $E_i \subseteq \mathcal{B}$. The provider chooses to disclose scores s_{ib} for $b \in D_i \subseteq E_i$. The receiver observes D_i and the scores, and must guess whether the remainder of the set represents withheld results, or non-existent results.

When endowment E_i is common knowledge, selective disclosure is dominated, as non-disclosure reveals the worst-case scenario and disclosure unravels with $D_i = E_i$ [Grossman, 1981, Milgrom, 1981]. This breaks down in at least two ways: (i) when E_i is common knowledge but disclosing results can be costly, leading to a cutoff strategy in s_{ib} [Verrecchia, 1983]; and (ii) when the receiver is unable to distinguish between non-existence and withholding, in that the sender can show evidence of having a result but not evidence of non-existence [Dye, 1985, Section 3].

In our setting an independent evaluator reports scores for cells that have been passed over by the provider, and each τ_b makes non-existence verifiable. Denote the available set and scored cells by $A_i = \{b : \tau_b < t_i\}$ and O_i; then

$$D_i \subseteq E_i \subseteq A_i, \qquad \theta = \Pr(b \notin D_i \mid b \in O_i \cap A_i)\,.$$

In general providers may have results which are not produced by the evaluator, and benchmarks may exist but not be run, so θ is calculated with reference to the set of cells covered by the evaluator, and not the actual endowment, leaving the Dye [1985] friction in place within A_i.

In practice, assessment of whether results have been withheld requires a notion of standing P_{ib}, since scores are not commensurable across benchmarks. The design requires that, conditional on innocent concerns, availability not be an argument in the determination of P_{ib} [Heckman, 1979]. A date rule cannot satisfy this, as $t_i - \tau_b$, t_i and τ_b are perfectly collinear. If any two of these arguments determine P_{ib}, then the remaining argument cannot be excluded from P_{ib} as a determining argument, which is the case here (Section 3; Appendix B bounds it).

Two consequences of this follow which need to be kept apart: (i) to measure θ, we require labels for disclosure, which we code in Section 5; and (ii) interpretation of estimates of θ as measures of withholding runs into A_i itself. Because availability is deterministically a step function of benchmark maturity, given by $\mathbf{1}[t_i - \tau_b > 0]$, no cells of equal maturity sit on opposite sides of the availability test. If standing is affected by maturity, then conditioning densely to make availability excludable violates positivity, which precludes global identification of selective disclosure using any metric of pool-relative standing, be it a win rate or percentile. Any scale for standing learned jointly will be subject to the same selection process which determines existence. Under assumptions of continuity, only the discontinuity at the threshold can be interpreted.

2.2 The independent score set

We work with a pinned version of Epoch's benchmarking hub scraped on August 17th, 2026. This data includes 819 model-versions. Since we are interested in release-level disclosure, we use organisation, model and release date to identify releases. Epoch uses a version attribute to distinguish different scaffolds within the same release, e.g. different reasoning effort or context length variants. Since the provider publishes one document for one release, we collapse these versions to releases, taking the maximum score among scaffolds and retaining the spread as a check. We end up with a dataset with 353 releases, 74 benchmarks and 3,082 release-benchmark cells that have an associated independent score. The data is clustered by provider.

2.3 Eligibility and benchmark dates

A provider cannot omit a benchmark that did not yet exist. To ensure temporal eligibility, we collect release dates for benchmarks: the earliest verifiable public release date (from Epoch if available, from the primary source of the benchmark if not, translating between preprint dates, code repository dates and different versions to get the first public release date). A cell is eligible only if the benchmark release date strictly predates the release's date and if the benchmark's release date is verified. This restriction assumes providers could not have access to unreleased benchmarks; if a provider was given early access to a benchmark, this assumption would not hold. Such violations shrink rather than create the reporting gap; they move available cells into the postdating category. Such violations leave no trace in our coded data: no document reports a benchmark before its public release date. This restriction leaves 2,355 out of the 3,082 cells eligible. Cells whose benchmark strictly postdates the release's date form a separate set of 476, cells that could not have been reported. We drop cells without a verified benchmark date rather than imputing a date; inaccuracies in dates shrink the boundary estimate we construct in Section 3. For comparing releases, we do not require a minimum number of benchmarks; this would exclude the releases that only have one eligible and one postdating cell. 57 out of 146 releases with both eligible and postdating cells have fewer than eight scored benchmarks. The more trivial obstacle, that how many benchmarks a release could report is set by when it shipped, is discussed in Appendix A.

To code disclosure, we adapt ORBIT, Outcome Reporting Bias In Trials. ORBIT is standardised in clinical trials, where disappearing outcomes (outcomes specified in the protocol but not reported in the publication) have a long literature on measurement [Chan et al., 2004, Kirkham et al., 2010, Dwan et al., 2013]. For existing cells, we record whether the provider reports a number, cites the benchmark without reporting, or reports a variant; for missing cells, we record whether a previous member of the same family reported it, if contemporaries reported it, or if no similar provider reported it. Of the 108 releases in the queue, 103 with eligible cells in our pinned dataset and five in a more recent vintage, we code the 100 for which we can locate a document from release time; 1,280 cells in total. None of the analysis before Section 5 depends on coded disclosure.

2.4 Measuring standing

Raw scores do not allow comparison across benchmarks, so we report relative performance as a percentile within benchmark by comparing each model to models whose release dates fall within 182 days before or after the focal release ("window"), using midranks for ties. We also report the number of peers and their share of the window released later than the focal model. This is the metric implied by a windowed comparison and what we demonstrate to be biased in a way that mimics selective disclosure.

2.5 Inference

Similar caution is needed when drawing inferences. Although there are 46 nominal provider clusters, 24 clusters provide fewer than ten cells, 20 are observed in only one release, and the largest five clusters contain 68.1% of the sample. We do not report analytic standard errors when there are fewer than twelve clusters, as the asymptotic distribution of cluster-robust standard errors can only be trusted above that number [Cameron et al., 2008]. Below that, the restricted wild cluster bootstrap with 9,999 replications is used for inference on the regression coefficients, imposing the null hypothesis and then resampling the Rademacher weights at the provider level. We use a cluster sign-flip randomisation test with 9,999 iterations to obtain p-values for the release-level means. The cluster sign-flip test reverses the sign of each organisation's contribution, accounting for the dependence within providers but not within benchmarks, which is accounted for in the two-way clustered regressions. Confidence intervals are obtained by percentile bootstraps with 10,000 resamples at the provider level. We use the $(k + 1)/(P + 1)$ formula to get p-values from both procedures, where k is the number of draws at least as extreme as the observed statistic and P is the total number of iterations. Since there are only 2^{G-1} distinct values of the statistic due to symmetry for G clusters, at least fifteen clusters are required to calculate a p-value with a precision of 0.0001; the 18 headline organisations clear it.

3 The Availability Gap

3.1 The placebo comparison

To separate withheld from non-existent, we require a control group of non-endowment [Dye, 1985]. Dates appear to provide one: a postdating benchmark could not be reported without prepublication access, whereas an eligible one could. A disclosure label would report whether it was, and none enters. Any innocent reason an eligible looks poor, difficulty drift, saturation, or measure change, operates equally well on both groups, meaning their difference should be zero. One such innocent reason is inherently asymmetric: a benchmark prior to shipping can lie in the training set for a given model and one after it cannot, so contamination increases eligible standing only, beyond any reweighting of the comparison.

Not zero. On average eligible beats postdating by 12.23 percentile points among 146 releases of both cells, with the median 10.69, and the proportion of releases where it wins is 78.1%. This computation does not know whether benchmarks are reported. By constructing one that treats both sides of the window equally (see below), we find eligible to beat by 9.14 percentile points across 141 releases, positive 70.9%, still without removing it. Using a cluster sign-flip randomization test of the difference at the 18 organisations with both cell types produces $p = 0.0001$ as the smallest available from the draws. Groups ineligible to be reported already stand lower than ones eligible, so it is not a control.

3.2 Decomposing the gap

In Table 1 we identify the failure using a placebo regression with an indicator for whether the benchmark postdates the release (always within release). With release effects alone this is −13.5 points. But benchmark fixed effects account for 63.4% of it, structurally since the placebo cells are constructed to exist primarily in those benchmarks arriving late, which are less saturated, with 8.3% of scores in the latter half of the 54 benchmarks with ≥ 10 scores in the top decile of its own range vs. 13.2% in the former ($p = 0.015$), leaving more room below the ceiling. The term capturing the asymmetry in peer-window, i.e. what share of comparisons are made to models released after the model of interest, controls for 39.8% on its own, and once we include both benchmark fixed effects and both window terms, the remaining is 0.003 points, with a confidence interval up to +3.4, which is no more than one third of the pre-control difference.

Conditioning set	Placebo coefficient	t	Absorbed
Release fixed effects only	−13.529 (2.038)	−6.637	0.0%
Release FE + benchmark fixed effects	−4.949 (2.183)	−2.267	63.4%
Release FE + peer-window asymmetry	−8.151 (1.865)	−4.370	39.8%
Release FE + peer count	−2.107 (1.882)	−1.120	84.4%
Two-way fixed effects + both window terms	−0.003 (1.748)	−0.002	100.0%

Table 1: Placebo coefficient by conditioning set, percentile points, within release. Two-way provider-by-benchmark standard errors (45 by 61 clusters; provider-only in Appendix B), n = 2,831. Absorbed is the fall in absolute value from row one; peer count suppresses; the last row has both.

For the second dimension, the mechanism is geometric, with the peer-window being symmetric in calendar time, 182 days before or after, but not model coverage, since coverage begins when the benchmark arrives. Figure 1 depicts both objects against benchmark maturity at release: the relative proportion of newer peers declines with maturity and jumps at the availability threshold, whereas standing runs through it without a detectable break. Thus, the mechanism is discontinuous where the outcome is not; composition is a property of timing alone.

The share of a cell's window released after the focal model is 0.4683 for eligible cells and 0.7065 for placebo cells, against 0.5 for a two-sided window. Within release, the slope of standing on that share is −30.63 percentile points per unit among eligible cells alone.

We plot standing versus the share of newer peers simultaneously for both types of cells in Figure 2. Instead of the two types of cells forming distinct point clouds, they follow a common downward trend that only appears after accounting for changes in the benchmark makeup (see Table 1); eligible and placebo cells lie at different points on that line. By construction, eligible and placebo cells are separated and never overlap on their assignment variable, the maturity of the benchmark at release; thus the difference between eligible and placebo cells is obtained by extrapolation rather than comparison on the common support. As a robustness check, we probe the continuity of the regression at the cutoff, the only part of the assignment variable where the analysis is identified, see Section 2.1 [Hahn et al., 2001]. The regression discontinuity analysis does not reveal a jump at the cutoff. Local linear regressions give negative coefficients for all bandwidths and orders, all confidence intervals include zero with the upper bound of the widest interval being +8.05, and the difference on a balance-driven bandwidth is −2.52 with a randomization p-value of 0.447 [Cattaneo et al., 2015]. The two types of cells differ in mean maturity by 793 days, and the discrepancy between them is driven by that distance rather than the cutoff point itself (Appendix B for the balance tests).

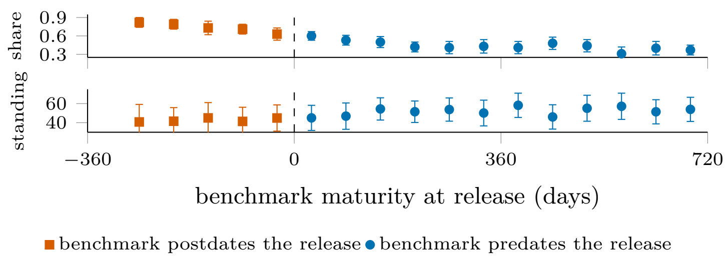

Figure 1: The mechanism and the outcome at the availability boundary. Top: the share of a cell's window released after the focal model, against benchmark maturity at release; it falls with maturity and jumps at the boundary. Bottom: within-benchmark standing on the same axis, continuous through it. Sixty-day bins; intervals clustered on organisation.

It's possible to decompose standing as a weighted combination of percentile-ranks against older and newer peers with the share of newer peers as weight, percentile $= (1 - w)\, P_{\text{old}} + w\, P_{\text{new}}$ (exact cell-wise). The side-balanced standing, setting the weight to one half, available for 92.5% of all cells (92.8% for eligible-or-placebo cells), cuts the same slope to −8.56.

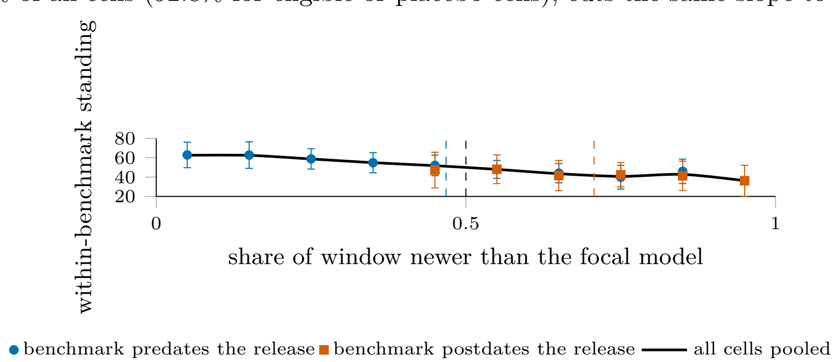

Figure 2: Standing against the share of a cell's window released after the focal model; dashed verticals mark group mean shares, the black dashed line the symmetric 0.5. Eligible and placebo cells lie on one declining curve, so the contrast between them is a position on the curve, not a fact about disclosure. Intervals clustered on organisation.

Despite the identity holding, the gradient is not a mere book-keeping tautology: permute the scores within the same benchmark, and the capability trend disappears while publishing dates, window membership, peer counts and newer shares are bit-wise identical. Under that null the slope has an average of −0.09 with a standard deviation of 3.12; the measured value of −30.63 is 9.8 standard deviations out, unmatched by any of the 300 draws. Further, fitting a linear regression within each benchmark of scores against publishing date, then reconstructing the scores as predicted values plus permuted residuals within each benchmark, meaning there's only a capability trend but no others, we are able to recover 93% of the slope over 200 draws. The comparison-set geometry by itself is inert; it only introduces bias when combined with the existing trend in capabilities.

3.3 Trimming and reweighting

What we describe here is an established estimation problem in nonparametrics: local means near a support boundary will be biased; the solution is to recentre, not censor them [Gasser and Müller, 1979, Fan, 1992]. The curious feature is that the window boundary coincides with the endowment: we had a yardstick to compare against a release on precisely those days where the model was not on the left boundary of that yardstick's coverage. What a provider could have held is defined by where the window is one-sided; there's no contrast left to exploit once we remove it. This coincidence is known in demography as the age-period-cohort identity: release date, benchmark vintage and the span between them are perfectly collinear, and each leaves an imprint on the data here: t_i (the capability trend), τ_b (saturation), and $t_i - \tau_b$ (window composition). In this setting, only the linear trends are unidentified [Fosse and Winship, 2019a,b], i.e. it's a slope problem. The slope is constrained by three sign restrictions: capabilities are increasing in calendar time, later benchmarks are less saturated, and standing increases with maturity. These are sign assumptions on each of the unidentified components; each is motivated by a measurement on some identified quantity. With those, the unidentified piece can explain at most 24 to 34% of the placebo gap across specifications, or roughly three percentile points; that's also where the saturated regression lands. The remainder is dominated by the identified reduced-form slopes; see Appendix B for a derivation.

Standing measure	Mean	Median	Positive	Sign-flip p	Releases	Cells
Windowed percentile, ±182d	+12.23	+10.69	78.1%	0.0001	146	100.0%
Rank against all models ever	+22.75	+22.66	89.0%	0.0001	146	100.0%
Trim to two-sided windows	+9.96	+10.69	72.2%	0.0001	115	70.4%
Side-balanced percentile	+9.14	+9.62	70.9%	0.0001	141	92.8%

Table 2: The label-free placebo null under each candidate measure: release-level mean standing of eligible minus postdating benchmarks, in percentile points. Every entry should be zero. Sign-flip p is the provider-clustered test of Section 2.5; Cells, the share of cells carrying the measure.

Table 2 shows the results. Ranking against all models ever scored almost doubles the contamination, while censoring to two-sided windows discards 30% of the cells to remove 19% of the gap (its row is the uncorrected gap on the subsample of two-sided cells, not a correction for it). Narrowing the window narrows the gap (down to +8.59 with 91 day windows), but always in proportion to how much asymmetry we've removed; it's not evidence of breaking the mechanism. Over four window sizes and four minimum benchmark counts, the gap is positive in all 16 specifications (from +8.59 to +21.66), correlating at $r = 0.991$ with the eligible-minus-placebo gap in window composition. Across leave-one-out folds of the 18 organisations, it ranges from +11.45 to +13.20. This is a covariate, not a subsample.

We don't see window composition as a control for a confound external to the comparison: it's a function of the same dates that define the indicator, so conditioning on it conditions the gap into its transmission channel. If a study wishes to hold disclosure labels to recover an estimate of θ, it must first establish that window composition is fixed prior to a provider deciding what to disclose, lest it control away part of what it intends to measure. Provider identity itself is a poor predictor of window composition (variance decomposition and uncorrected coverage tilt in Appendix B); the switch to side-balanced percentiles is the only change in measure that makes a meaningful dent in the gap (reducing the placebo mean to +9.14 and the gradient in the eligible cells to −8.56); see Figure 4 in Appendix B for a plot against the uncorrected measure, close to flat where cells are dense. It can't fix what remains because while balancing controls for how much weight is on either side of a model, it can't adjust for which models an evaluator decided to grade early, and that remains a confound. It is undefined where a side has no cells, and remains an accompaniment rather than a replacement for the uncorrected measure.

4 Evidence from HELM

All of this has been on a percentile computed here, on a panel composed here. It is possible that only the percentile design has this property, and not leaderboards generally. For example, HELM Lite is a leaderboard that publishes all its accuracy data at each versioned release, and retains this history. It also publishes a pool-relative headline [Liang et al., 2023], so the same evidence is read at 14 points in time while the pool expands.

When a leaderboard drifts, the first question is: has the suite changed? From v1.0.0 to v1.13.0, the 10 scenario columns are unchanged: no scenarios added, removed or renamed, except for a single column header relabel. Among the 24 models that span all 14 releases, all 240 published scenario scores remain bit-identical, as the evaluation pool expands from 30 to 91 models.

Published Mean win rate for all 24 constant models drifts (mean $|\text{drift}| = 0.1232$, max $|\text{drift}| = 0.2178$ for Mixtral 8x7B). This drift is mostly arithmetic: Mean win rate is computed as a win fraction, and a constant model compares against a larger, later pool over time; this drift correlates with initial standing ($r = -0.766$).

The key drift is in another direction. If the drift was a monotonic function of the bit-identical evidence then it would not change the ranking: monotonic transformations preserve rank order. This is not the case: 22 of the 24 constant models drift lower while two drift higher, and 5 of 276 pairs (1.8%) reorder between the first and last version; all are published rank inversions under bit-identical evidence. Figure 3, in Appendix B, plots the rank of each of the 24 constant models at the first version against the last. Each rank reversal is a crossing, visible and countable, on a bit-identical evidence base. This is distinct from overfitting to a test set [Roelofs et al., 2019]: here the evidence is bit-identical. It is the reference set that drifts. We observe a similar drift, from another cause, in Singh et al. [2025]: deprecation policy rather than a coverage boundary changing which models remain available; in both cases, the evidence remains constant and the ranking drifts with the reference set.

We also track a control: the HELM Capabilities leaderboard is from the same organisation, whose maintainers have diagnosed the win rate as pool-dependent [Xu et al., 2025]; it operates on the same infrastructure and release cadence, with a comparably expanding pool ($22 \to 68$ models in 16 versions). Instead of reporting the win fraction, its headline metric is an absolute mean over scenarios. Among 22 models that span all versions, all 110 scenario scores remain bit-identical, none of the 22 headline values drift, and there are no rank reversals, at the endpoints or at any time. Because the headline is an absolute mean over bit-identical scores on a fixed suite, it cannot drift with the pool: no drift is arithmetic, a scope condition rather than a test that could have failed.

None of this needs to be interpreted as a mistake on the part of HELM: their absolute headline, on their own Capabilities leaderboard, is the point. A pool-relative headline is a joint statement, about the model and about the population assessed beside it; read later as statements about the models alone, two such numbers move standing between providers for reasons belonging to neither. The problem of making different-time metrics comparable is a classical equating problem [Kolen and Brennan, 2014], which is bypassed by estimating a scale jointly [Ho et al., 2025]. The panel and the leaderboard fail identically: the reference set, not the evidence, moves the number.

5 Coding the Record

Possible reasons for absence from model cards include irrelevancy to the model, absence of running, or withholding after running. Of these only the latter qualifies as disclosure [Grossman, 1981, Milgrom, 1981, Dye, 1985], and is thus one readers can price. The within-family drop narrows this: if the provider reported b at t but not at $t+1$, and an independent score exists, then (i) b is relevant, (ii) the provider had access to the benchmark harness, and (iii) the independent score serves as a registry-like counterfactual [Chan et al., 2004, Dwan et al., 2013, Kirkham et al., 2010]. But this does not establish that b was run for $t+1$, narrowing the Dye [1985] uncertainty but not closing it.

One coder coded the document released at release time for each of the 100 releases we can locate documents for (8 of 108 queued have none). We record which benchmarks they report, variants, mentions without scores, and explicitly-benign non-reports; we use these primitives and family history to mechanically derive the ORBIT categories. This results in 1,280 coded cells, 31 of which come from five of the most recent releases not in our pinned panel and are excluded. For a reliability study, a different coder double-coded a provider-stratified random fifth of the releases blind to the first and with no calibration communication (Appendix B notes the one release where this blinding was broken). Kappa scores are for the derived cells [Cohen, 1960], confidence intervals by resampling releases. Coders agree on the raw outcome (was the benchmark reported or not) at 0.95 $[0.80, 1.00]$, all categories at 0.85 $[0.72, 0.95]$, and high/low suspicion categories at 0.81 $[0.48, 1.00]$. Coders' E cell output is identical (promising, but the category is too sparse for a reliability check; confusion structure in Appendix B). Among the 1,249 coded cells eligible for our panel, 76.9% show no reported number; the mean over releases is 73.1% $[65.9, 79.2]$. (This is a descriptive rate, not a test of selection.)

Standing measure	Coded gap	95% CI	No labels	Difference
Windowed percentile, ±182d	+11.40	[+5.6, +17.0]	+14.13	−2.73
Side-balanced percentile, ±182d	+7.90	[−0.4, +15.2]	+9.82	−1.92
Rank against all models ever	+22.33	[+18.0, +28.1]	+24.50	−2.17
Windowed percentile, ±90d	+8.15	[+1.3, +14.7]	+10.36	−2.21
Windowed percentile, ±365d	+18.03	[+13.9, +23.1]	+20.53	−2.50

Table 3: The coded omission deficit beside the label-free gap, percentile points, over 71 releases and 13 providers (70 side-balanced). No labels is the contrast with no disclosure coding; Difference is Coded gap minus No labels. Provider-clustered bootstrap intervals; four of five exclude zero.

Table 3 is the output of this section. Reporting vs. non-reporting: +8.7 points $[4.1, 13.9]$ difference in 63 releases. The intended availability contrast (non-reported eligible vs. post-dating) is +11.4 $[5.6, 17.0]$ in 71 releases where both are present; a null hypothesis of zero is rejected at $p = 0.003$. Read against Section 3, this is the availability gap: the original label-agnostic difference on those same releases is +14.1 and the labelled difference is lower by 1.9 to 2.7 points on all measures and time windows. This is strictly an accounting identity: it equals +2.7, the average over those same releases of the fraction reported × the reported-vs-not-reported gap within the release; our labelling exercise can only influence this contrast through composition.

The residual is composition: the +8.7 points standing difference between reported and non-reported eligible benchmarks within release is significant against 9,999 label shuffles that preserve category counts ($p = 0.0001$, bounded by the sample of shuffles, null st. dev. 1.9) and cannot be driven by the availability gap. It is also stable to coding mistakes, re-classifying variants, and jackknifing (same as the availability gap); and stable to the identity of the benchmarks (fixed effects of benchmarks in addition to releases raise the coefficient from +7.4 to +9.3; adding the window-share control retains +7.4). What it can't do is commit to the causal interpretation that standing drove reporting (relevance, salience, contamination within eligible, etc. can sway both reporting and standing).

The remaining distance is possession; the original drop estimator and four possession splits agree. The drop estimator realises 16 drops across 15 releases and finds +2.9, against the prediction of withholding; $p = 0.25$ on sign flip across five providers. In 295 cells documenting a run, none claim to have run the benchmark without reporting a score or outcome. A change-based test (each drop against its standing at the previous release) finds −3.7, consistent with the prediction, permutation $p = 0.16$ across 42 transitions, no additional signal in possession tiers or same-sized substitutions. Details and three unrepaired caveats in Appendix B.

6 Conclusion

The coded record returns the number and the reason to distrust it: +11.4 points, rejecting zero yet short of its label-free counterpart by what composition implies, too imprecise to exclude a strategic component. Such a study turns composition into apparent concealment. Three claims separate: non-reporting is extensive, composition is performance-related, withholding not established, the possession tests returning null, absent, or opposite-signed evidence. Growing the 16-drop record is a reporting-standard question: a joint scale [Ho et al., 2025] shrinks but does not close the placebo gap, while a suite carried across releases would make later silence informative, narrowing the Dye [1985] pooling.

References

Andrew M. Bean, Ryan Othniel Kearns, Angelika Romanou, Franziska Sofia Hafner, Harry Mayne, et al. Measuring what matters: Construct validity in large language model benchmarks. In Advances in Neural Information Processing Systems 38, 2025. arXiv:2511.04703.

Benjamin B. Bederson, Ginger Zhe Jin, Phillip Leslie, Alexander J. Quinn, and Ben Zou. Incomplete disclosure: Evidence of signaling and countersignaling. American Economic Journal: Microeconomics, 10(1):41–66, 2018. doi: 10.1257/mic.20150178.

A. Colin Cameron, Jonah B. Gelbach, and Douglas L. Miller. Bootstrap-based improvements for inference with clustered errors. Review of Economics and Statistics, 90(3):414–427, 2008. doi: 10.1162/rest.90.3.414.

Matias D. Cattaneo, Brigham R. Frandsen, and Rocío Titiunik. Randomization inference in the regression discontinuity design: An application to party advantages in the U.S. senate. Journal of Causal Inference, 3(1):1–24, 2015. doi: 10.1515/jci-2013-0010.

An-Wen Chan, Asbjørn Hróbjartsson, Mette T. Haahr, Peter C. Gøtzsche, and Douglas G. Altman. Empirical evidence for selective reporting of outcomes in randomized trials. JAMA, 291(20):2457–2465, 2004. doi: 10.1001/jama.291.20.2457.

Jacob Cohen. A coefficient of agreement for nominal scales. Educational and Psychological Measurement, 20(1):37–46, 1960.

Yifan Dai, Drew Fudenberg, and Harry Pei. Bayesian persuasion with selective disclosure, 2026. arXiv:2601.05914.

David Dranove and Ginger Zhe Jin. Quality disclosure and certification: Theory and practice. Journal of Economic Literature, 48(4):935–963, 2010. doi: 10.1257/jel.48.4.935.

Kerry Dwan, Carrol Gamble, Paula R. Williamson, and Jamie J. Kirkham. Systematic review of the empirical evidence of study publication bias and outcome reporting bias. PLoS ONE, 8(7):e66844, 2013. doi: 10.1371/journal.pone.0066844.

Ronald A. Dye. Disclosure of nonproprietary information. Journal of Accounting Research, 23(1):123–145, 1985. doi: 10.2307/2490910.

Epoch AI. Ai benchmarking hub. https://epoch.ai/benchmarks, 2026. Data snapshot of 17 August 2026, CC-BY.

Jianqing Fan. Design-adaptive nonparametric regression. Journal of the American Statistical Association, 87(420):998–1004, 1992. doi: 10.1080/01621459.1992.10476255.

Ethan Fosse and Christopher Winship. Bounding analyses of age-period-cohort effects. Demography, 56(5):1975–2004, 2019a. doi: 10.1007/s13524-019-00801-6.

Ethan Fosse and Christopher Winship. Analyzing age-period-cohort data: A review and critique. Annual Review of Sociology, 45:467–492, 2019b. doi: 10.1146/annurev-soc-073018-022616.

Theo Gasser and Hans-Georg Müller. Kernel estimation of regression functions. In Smoothing Techniques for Curve Estimation, volume 757 of Lecture Notes in Mathematics, pages 23–68. Springer, 1979. doi: 10.1007/BFb0098489.

Avijit Ghosh, Anka Reuel, Jenny Chim, Wm. Matthew Kennedy, et al. Evaluation cards: An interpretive layer for ai evaluation reporting, 2026. arXiv:2606.09809.

Sanford J. Grossman. The informational role of warranties and private disclosure about product quality. Journal of Law and Economics, 24(3):461–483, 1981. doi: 10.1086/466995.

Jinyong Hahn, Petra Todd, and Wilbert Van der Klaauw. Identification and estimation of treatment effects with a regression-discontinuity design. Econometrica, 69(1):201–209, 2001. doi: 10.1111/1468-0262.00183.

James J. Heckman. Sample selection bias as a specification error. Econometrica, 47(1): 153–161, 1979. doi: 10.2307/1912352.

Anson Ho, Jean-Stanislas Denain, David Atanasov, Samuel Albanie, and Rohin Shah. A rosetta stone for AI benchmarks, 2025. arXiv:2512.00193.

V. Joseph Hotz and Mo Xiao. Strategic information disclosure: The case of multiattribute products with heterogeneous consumers. Economic Inquiry, 51(1):865–881, 2013. doi: 10.1111/j.1465-7295.2010.00340.x.

Ginger Zhe Jin and Phillip Leslie. The effect of information on product quality: Evidence from restaurant hygiene grade cards. Quarterly Journal of Economics, 118(2):409–451, 2003. doi: 10.1162/003355303321675428.

Ginger Zhe Jin, Michael Luca, and Daniel Martin. Is no news (perceived as) bad news? an experimental investigation of information disclosure. American Economic Journal: Microeconomics, 13(2):141–173, 2021. doi: 10.1257/mic.20180217.

Woon-Oh Jung and Young K. Kwon. Disclosure when the market is unsure of information endowment of managers. Journal of Accounting Research, 26(1):146–153, 1988.

Jamie J. Kirkham, Kerry M. Dwan, Douglas G. Altman, Carrol Gamble, Susanna Dodd, Rebecca Smyth, and Paula R. Williamson. The impact of outcome reporting bias in randomised controlled trials on a cohort of systematic reviews. BMJ, 340:c365, 2010. doi: 10.1136/bmj.c365.

Michael J. Kolen and Robert L. Brennan. Test Equating, Scaling, and Linking: Methods and Practices. Springer, 3rd edition, 2014. doi: 10.1007/978-1-4939-0317-7.

Blaine Kuehnert, Nari Johnson, Ravit Dotan, and Hoda Heidari. Disclosure or marketing? analyzing the efficacy of vendor self-reports for vetting public-sector ai. In ACM Conference on Fairness, Accountability, and Transparency (FAccT), 2026. doi: 10.1145/3805689.3812220.

Percy Liang, Rishi Bommasani, Tony Lee, et al. Holistic evaluation of language models. Transactions on Machine Learning Research, 2023. arXiv:2211.09110.

Wei Ma and George H. Chen. Missing not at random in matrix completion: The effectiveness of estimating missingness probabilities under a low nuclear norm assumption. In Advances in Neural Information Processing Systems 32, 2019. arXiv:1910.12774.

Justin McCrary. Manipulation of the running variable in the regression discontinuity design: A density test. Journal of Econometrics, 142(2):698–714, 2008. doi: 10.1016/j.jeconom.2007.05.005.

Paul R. Milgrom. Good news and bad news: Representation theorems and applications. Bell Journal of Economics, 12(2):380–391, 1981. doi: 10.2307/3003562.

Rebecca Roelofs, Vaishaal Shankar, Benjamin Recht, Sara Fridovich-Keil, Moritz Hardt, John Miller, and Ludwig Schmidt. A meta-analysis of overfitting in machine learning. In Advances in Neural Information Processing Systems 32, 2019.

Shivalika Singh, Yiyang Nan, Alex Wang, Daniel D'Souza, Sayash Kapoor, Ahmet Üstün, Sanmi Koyejo, Yuntian Deng, Shayne Longpre, Noah A. Smith, Beyza Ermis, Marzieh Fadaee, and Sara Hooker. The leaderboard illusion. In Advances in Neural Information Processing Systems 38, Datasets and Benchmarks Track, 2025. arXiv:2504.20879.

Sandip Sinharay and Paul W. Holland. The missing data assumptions of the NEAT design and their implications for test equating. Psychometrika, 75(2):309–327, 2010. doi: 10.1007/s11336-010-9156-6.

Leon Staufer, Mick Yang, Anka Reuel, and Stephen Casper. Audit cards: Contextualizing ai evaluations, 2025. arXiv:2504.13839.

Sang Truong, Serena Wang, Nick Haber, and Sanmi Koyejo. Strategic evaluation: Incentivizing AI capability coverage with private benchmarks, 2026. Working paper.

Robert E. Verrecchia. Discretionary disclosure. Journal of Accounting and Economics, 5: 179–194, 1983. doi: 10.1016/0165-4101(83)90011-3.

Jialiang Xu, Yifan Mai, and Percy Liang. HELM capabilities: Evaluating LMs capability by capability. Stanford Center for Research on Foundation Models blog, 2025. URL https://crfm.stanford.edu/2025/03/20/helm-capabilities.html.

Chuyifei Zhang, Hongyu Cui, Xiaowen Huang, and Jitao Sang. Positional failures in long-context LLMs: A blind spot in reasoning benchmarks, 2026. arXiv:2605.23170.

A What a Provider Could Report

The comparisons above do not take into account that the set of results a provider could publish is increasing as new benchmarks are published. In our panel the number of dated benchmarks increases from 11 at the end of 2019 to 62 by 2026. Once a new benchmark appears it remains available to every release after it. In this sense, the number of possibilities is always weakly increasing.

At the same time, what a single release could report grows more slowly and less evenly. On average, by half-year, the number of eligible benchmarks per release is 6.8 in 2023 H1, declining to 3.9 in 2024 H1 as releases outpaced the instruments now used to score them, and it is 10.1 by 2026 H1. Thus, in our data, the two series move together over the sample and apart within it. This lends some evidence that the opportunity set may partially be shaped by the date of the releases and not by the providers.

This simple observation shows that a count of omissions is merely a count of the calendar. If reported tables do not expand one for one with the stock, then omission rates will increase by arithmetic forces alone, and comparisons between early and late releases are a comparison across eras, not across providers. All that remains is composition: which benchmarks providers drop, conditional on availability, compared to what an indifferent selection over the same set would have produced. To conduct this exercise it is necessary to know what was available. But once that is known it is also necessary to know how the release stood on each available benchmark. As we show in the rest of the paper, the latter is far more difficult than the former [Dranove and Jin, 2010].

B Additional Checks

Does the provider choose the window composition? Section 3.3 conditions on window composition, which is admissible only if this is settled before the reporting decision. Fitting a regression of the fraction of a cell's window released after the focal model, on provider dummies, yields an R^2 of 0.071 (45 organisations); for comparison, fitting benchmark dummies yields R^2 of 0.002 (61 benchmarks), while the release date alone explains R^2 of 0.104. At the level of cells, window composition appears almost idiosyncratic. This is consistent with it being determined by the scoring schedule of the evaluator, as opposed to the provider. It is not enough to establish it.

Is the evaluator's coverage selective? The estimand also conditions on which cells are scored by the evaluator. This factor is by construction upstream of all others. On average, a release has 8.7 benchmarks scored; this varies with standard deviation 8.0, and spans a range of 1–38. It correlates with the provider's release count ($r = 0.203$) and with release date ($r = 0.265$), where releases from the densest quartile of providers average 9.6 scored benchmarks, versus 8.4 for the rest. This suggests a mild tilt toward larger and later providers, which we do not adjust for in this paper.

Bounding the unidentified component. Let t be the release date, τ be the benchmark vintage, and $a = t - \tau$ be the maturity of the benchmark when the model met it. A regression of standing on these terms is by construction singular (the age-period-cohort problem), and only identifies $\Pi_p \equiv \beta_a + \beta_t$ and $\Pi_c \equiv \beta_\tau - \beta_a$, leaving a one-parameter family indexed by $\lambda \equiv \beta_a$. Fitting the reduced form across the 2,831 comparable cells, we obtain $\Pi_p = +4.12$ and $\Pi_c = -1.36$ percentile points per year. We can rely on three restrictions, derived from the body of the paper itself as opposed to external assumptions. (1) Capability is increasing in calendar time, i.e. $\beta_t \geq 0$. (2) Later benchmarks are less saturated, as directly quantified in Section 3.2, i.e. $\beta_\tau \leq 0$. (3) Standing is increasing in maturity, since a model on a young benchmark is ranked against newer peers, i.e. $\beta_a \geq 0$. Taken together, this means that $0 \leq \lambda \leq \min(\Pi_p, -\Pi_c) = 1.36$, where it is saturation, not capability, which is the binding restriction. On average, eligible and placebo cells differ by 2.17 years in maturity. Thus, at most, this unidentified component explains $1.36 \times 2.17 = 2.94$ percentile points, or 24% of the observed gap. If we include centred squared and cubed terms in t and τ (which are not collinear with a), then the ceiling moves to 34% and 31%. If we shuffle scores within benchmark, preserving every date and the rank deficiency while destroying the capability trend, then this quantity drops to an average of 0.02 percentile points across 30 draws, peaking at 0.24, confirming that the bound reflects the trend the geometry multiplies rather than the geometry alone.

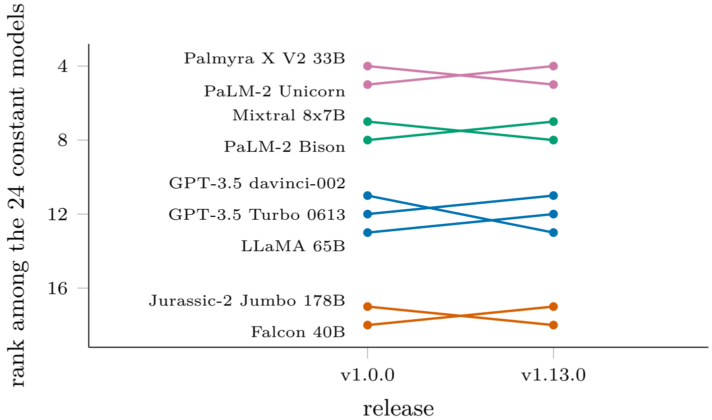

Figure 3: The nine HELM Lite models whose published order changed between the first release and the last, ranked among the 24 whose 240 scenario scores are bit-identical throughout; coloured by group of mutually reversing models, four groups and five reversed pairs in all. The control: across 16 HELM Capabilities releases behind an absolute headline, with the pool growing from 22 to 68, none of the 22 constant models' headline values moved and no pair reversed.

The cutoff is a discontinuity. Treatment eligibility $\mathbf{1}[t_i > \tau_b]$, being a deterministic function of dates, is the problem that global approaches confront. Equating across different instruments requires a common conditional relation across samples, which samples split by a cutoff do not satisfy [Sinharay and Holland, 2010]; weighting by observed cell probability assumes a non-vanishing observed probability, which a deterministic cutoff rules out [Ma and Chen, 2019]; finally, a deterministic cutoff is a sharp discontinuity so that, at the cutoff, standing is identified under continuity conditions regardless of what is true elsewhere [Hahn et al., 2001]. We estimate local linear regressions using a triangular kernel and errors clustered at the provider level with results of -8.82 (4.76) at ± 60 days, -6.85 (4.09) at ± 90, -4.08 (3.11) at ± 180 and -2.85 (2.35) at ± 250; adding a quadratic term at each bandwidth does not take any of these estimates across zero. All eight estimates are negative; every confidence interval includes zero and the highest upper limit of the CI is $+8.05$, strictly below the $+12.23$ the global comparison reports. Consistent with local randomization inference at the cutoff [Cattaneo et al., 2015], we also increase symmetric bandwidths around the cutoff and evaluate balance of the release's benchmark count, scaffold count, and scaffold spread; the minimum p value over these tests is 0.335 at ± 20 days and 0.571 at ± 40, decreasing to 0.058 at ± 60; we pick ± 40 days, ad hoc based on the balance tests, and thus the randomization p value reported below is not corrected for that choice. The same difference is -5.96 ($p = 0.23$) at ± 20 days and $+0.43$ ($p = 0.88$) at ± 60. With 113 postdating and 114 eligible cells in this range, the difference in mean standing is -2.52, randomization p of 0.447. There is no bunching against the cutoff: the ratio of cell counts within 30 days after vs. 30 days before the cutoff is 0.95 (80 vs. 84 cells) [McCrary, 2008]. The mechanism is discontinuous where standing is not. At the cutoff, the fraction of a cell's window released after the focal model has a step size of $+0.083$, translating, using the -30.63 gradient for eligible cells, to about -2.5 percentile points, comfortably within all confidence intervals above, so the cutoff design discriminates concealment from the availability gap only above that size at the cutoff; at the narrowest bandwidths, the full -12.23 is not rejected either. Peer count and raw score are continuous. The global difference arises because the average cell maturity differs between the two samples by 793 days, not from anything at the cutoff.

A standing measure with no comparison window. Section 4 points at a scale estimated jointly across benchmarks rather than against contemporaries [Ho et al., 2025], which eliminates the comparison window from the measure. Fitting one does not close the gap. We jointly estimate ability and difficulty across the 63 benchmarks reported on a single zero-one scale, with a logit link, and regularise the loading toward one (unregularised, with this data density, the loading is unidentified and goes above 65,000). Among benchmarks that carry at least twenty releases, and releases that carry at least three benchmarks, there are 31 benchmarks, with a total of 1,669 eligible cells. We fit the model on eligible cells alone, making the 226 postdating cells it scores out-of-sample.

The eligible-minus-placebo gap in the residuals is +0.336 residual SDs versus +0.449 for the windowed percentile on the same contrast, at $p = 0.005$, and positive in 64.0% of the 89 releases holding both kinds. Thus, removing the window shrinks the gap, but by only about a quarter, leaving it positive, which places this approach, on its jointly estimated scale, beside the 19% trimming and 25% side-balancing approaches, on their own scales, rather than ahead of them. Part of this gap may be the same selection re-expressed, since the difficulty parameters are estimated from the maturity-selected cells that exist; it also builds in an asymmetry, since its placebo cells are out-of-sample and every eligible cell is in-sample. When we shuffle the scores within a benchmark, preserving every date and the rank deficiency, this measure falls to an average of 0.02 over 30 draws.

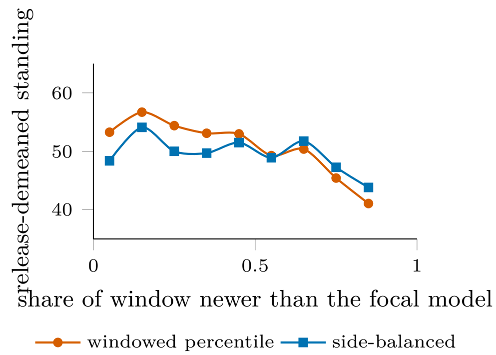

Figure 4: The side-balanced percentile against the windowed one, on eligible cells within release. The windowed measure slopes across the range while the side-balanced one is close to flat where cells are dense; the quoted gradients, −30.63 to −8.56 per unit share, are the within-release regression slopes on these cells, and the release-demeaned binned means shown are correspondingly flatter.

We also check whether the positive gap is a product of looking forward. A backward-only standing, ranking only among older peers, removes the window's forward half altogether, and still yields a label-free gap of +12.11 across 142 releases, eliminating this explanation.

Drop-design ceiling and calibration. The panel contains 144 pairs of adjacent same-family releases, spanning 711 shared eligible benchmarks and 15 organisations. Limiting to pairs with at least eight scored benchmarks for both releases narrows it to 37 pairs, 482 shared benchmarks and 8 organisations (the effective number of clusters), and we are in the regime where Section 2.5 reports no analytic standard error. Over 400 draws from each of the 117 releases that qualify for this test, the standard deviation of a drop gap under the null of no selective withholding is 19.4 percentile points when one benchmark is dropped, 14.4 when two are, and 12.4 when three are. At standard two-sided levels a single-drop event must therefore move the gap by roughly 38 points before it separates from noise, three times the contamination itself, and no single drop can be read alone.

Evidence of possession. Withholding argues that providers selectively choose which retained results not to report; the disclosure measure registers only what is printed. Four slices put reporting into the perspective of possession. (I) In the direct measures, a report must say that a benchmark has been run but not provide a measure or result; no such direct withholding report exists across 295 documented runs: all six Named No Number cases provide a result in a figure, chart or comparison. (II) Omission deficit by evidence of possession. Withholding argues that omission should attenuate and reverse with increased evidence of possession; composition says no. The deficit averages 18.9 (21 releases, dropped); 11.3 (51 releases, peers report); and 11.3 (69 releases, silent). There is no attenuation (more negative deficit) with increased evidence of possession: instead, the deficit is highest for dropped, the opposite gradient of the signature. (III) Drop tests on change rather than level. The previous estimator tests levels; a more stringent test would be if providers drop a benchmark when the family has declined on it, netting out structural salience and relevance. The average change in 42 family transitions with a scored prior report is −3.7 percentile points

(33 dropped and 86 retained cases), in the expected direction for withholding and $p = 0.16$ with a count-preserving permutation within transitions; in the original scores (kept where on the unit interval) the difference is -0.7 points and $p = 0.71$. These are two different subsamples of the 63 derived drop cells: the level test retains the 16 in releases that also retain a minimum of two reported benchmarks for the comparison; the change test retains 33 with an independently scored prior and post, with no minimum retention requirement. (IV) Substitutions. In the extreme case that withholding only argues for a constant reporting table, providers must drop a benchmark and report another in its place. No such cases are found in the record. None of these point to withholding. Only the change-based difference is in the predicted direction and cannot be statistically distinguished from the null of permutation; all others are null, non-existent, or sign-reversed.

Reliability. The second coder coded 20 of the 100 readable releases, stratified by provider. Categories that import peers' evidence are dependent across releases, a dependence not modelled by resampling releases. In 15 of these, 194 cells pair; excluded are i) one release with zero direct disclosure, whose rows for missing evidence are dropped by the pairing, a conservative selection as all eleven cells are trivial agreement, and ii) four releases not in the pinned sample. The reporting/not-reporting contingency table crosses in only one direction (47 reported/reported and 143 not/not): none the first coder reported was scored differently by the second; and all four disagreements with reporting switched are in a single release with a note to the second coder retaining four scores disputed by the first (in this case, the release document breaks its own blindness). Without it, agreement on reported/not is perfect, with an overall κ of 0.88 on 182 cells. The 14 disagreements are mostly (7) about the borderline between C and A, one between mention and scored, two about the derived category G/H split in a single release, and four are above. Three record limits are known but not corrected: eight releases cannot be located, arguably non-random, and likely to report less; the cells scored by the evaluator are subject to selection by provider emphasis, arguably biasing the reporting record up; and the take-best across scaffolds scoring does not explore asymmetric behaviour around the boundary. Full agreement on the high-suspicion marginals (5.7% each) pushes the split category κ up to 0.81 despite rarity; the large confidence interval is caused by eleven cells concentrated in a few releases.

Provider-only clustering. For comparison, the paper clusters by provider and benchmark in regressions. Errors with only provider clustering are smaller: 1.271 versus 2.038 by release effects, ratio 1.60; and 1.652 versus 1.748 in the fully saturated case, ratio 1.06. This says that there is genuine autocorrelation in the raw difference by benchmark, and not much in the benchmark effects and window terms, as suggested by the decomposition. While the headline test at $p = 0.0001$ for sign flip retains the provider dependence, it is the row above in the two-way regressions which states that the contamination is present in both dimensions, at $t = -6.6$.

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   Answer: [Yes]

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    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [Yes]

    Justification: The released code documents how each number is produced, the snapshot manifest records the exact data build, and the frozen external evidence records a source URL, a retrieval date and a SHA-256 for every file.

14. Crowdsourcing and research with human subjects

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [N/A]

    Justification: The work involves no crowdsourcing and no human subjects.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [N/A]

    Justification: The work involves no human subjects, so no review board approval was required.

16. Declaration of LLM usage

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does not impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

    Answer: [Yes]

    Justification: All disclosure-coding categories standing in the paper were hand-assigned by a human coder from archived release-time documents under the frozen protocol, with an independent second coder recoding a provider-stratified 20% sample (Section 5), so no number reported here depends on a model's judgment. A superseded early pass had used a large language model on the same task; those labels were cleared before analysis and none enters. Every number is produced by the released code from the pinned snapshot. Language models were otherwise used only for writing and code assistance.